

Modeling interception movement distributions

Seman Libbiss and Paz Flashner

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Jason Friedman and Moni Shahr

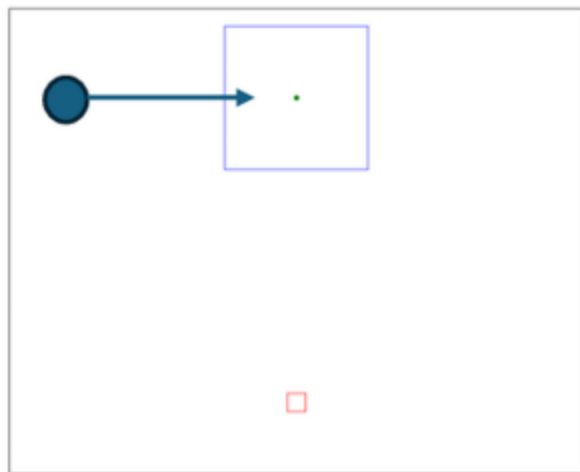
A compact personal representation for stochastic movement generation

The task and the evaluation split

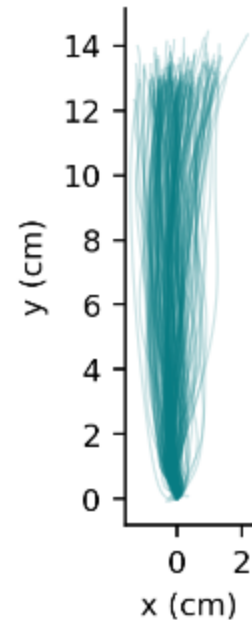
4,732 trials from 28 participants

Four rotations: 17 train / 4 validation / 7 test

Task display



subject01: 180 recorded trials



Free eye movement. All 180 trajectories from the first participant by identifier. Median retained trials: 179.5.

The compression pipelines

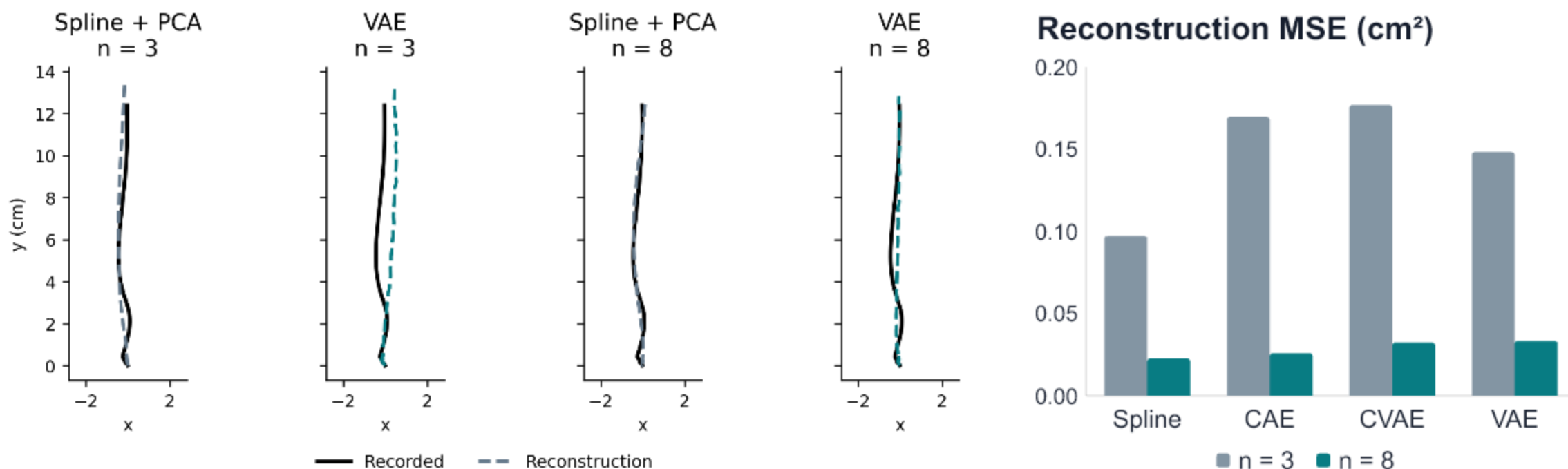
Every trajectory becomes 100 x 2 positions on normalized time.

Stage	Spline + PCA	Neural models
Input	100 x 2 positions	200 standardized coordinates
Encoder	9 coefficients per axis; PCA on N x 18 matrix	Two 256-unit layers; n-value code
Decoder	Inverse PCA; two spline basis products	Two 256-unit layers; 200 coordinates
Timing output	Ridge on code + 5 conditions	Separate two-duration head

VAE: variational autoencoder. CVAE adds task conditions; CAE uses a deterministic code. Matched n = 3 and 8.

Reconstruction favors spline + PCA

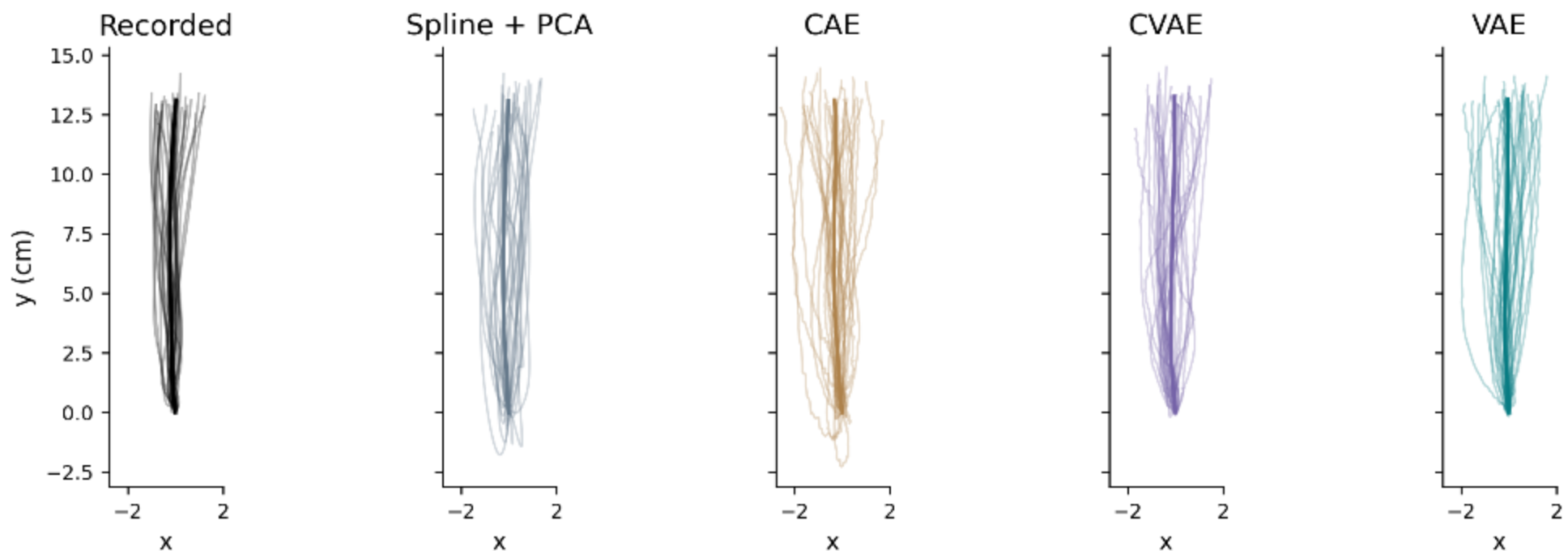
The encoder receives the observed test trajectory.



Black: recorded; dashed: reconstructed. One median-duration trial. Lower MSE is better. Spline beats each neural model under BH-140; spline vs CAE $n=8$ does not survive Holm.

Generation uses separate context trials

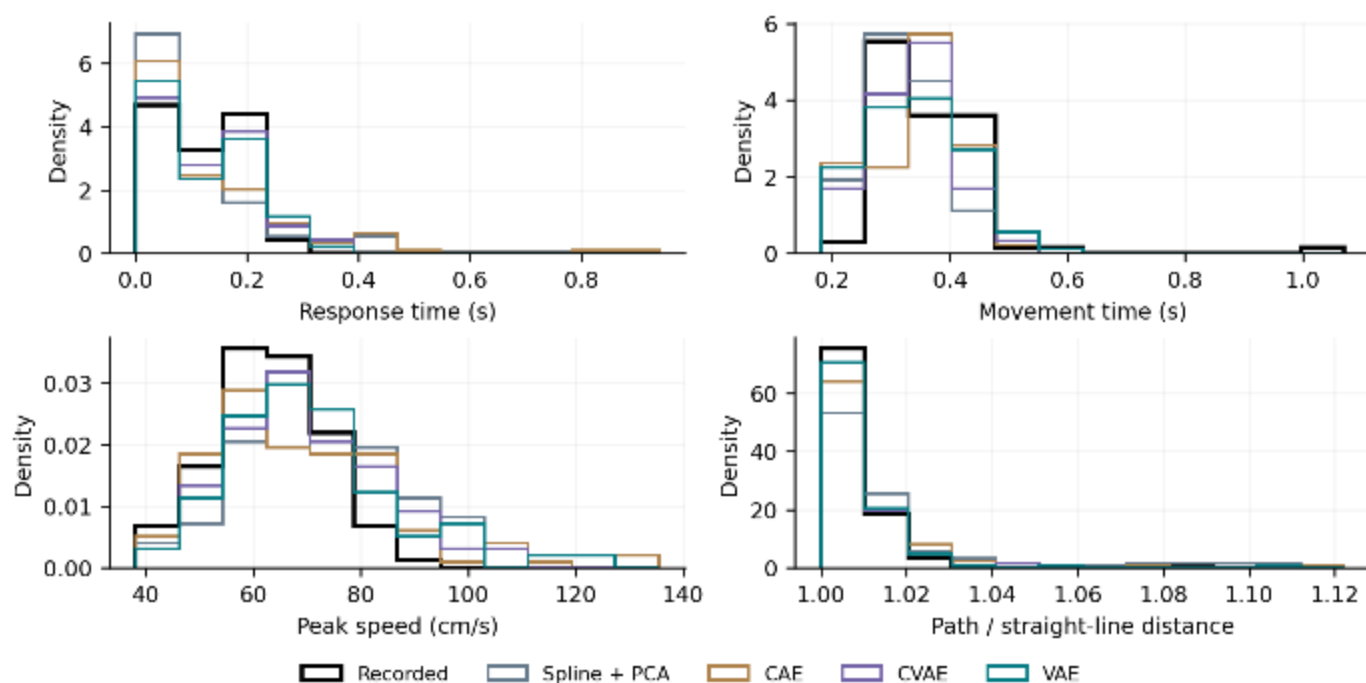
Context codes establish a personal center. We decode 120 draws around it using shared training covariance.



Same participant, separate query recordings. $n=8$, all models. Both capacities appear in the report. Thin: first 30 paths. Thick: mean of all paths. Equal x/y scale in cm.

Generated kinematic-feature distributions

Histograms show timing, speed and curvature. KS summarizes agreement across eleven features.



Mean feature KS

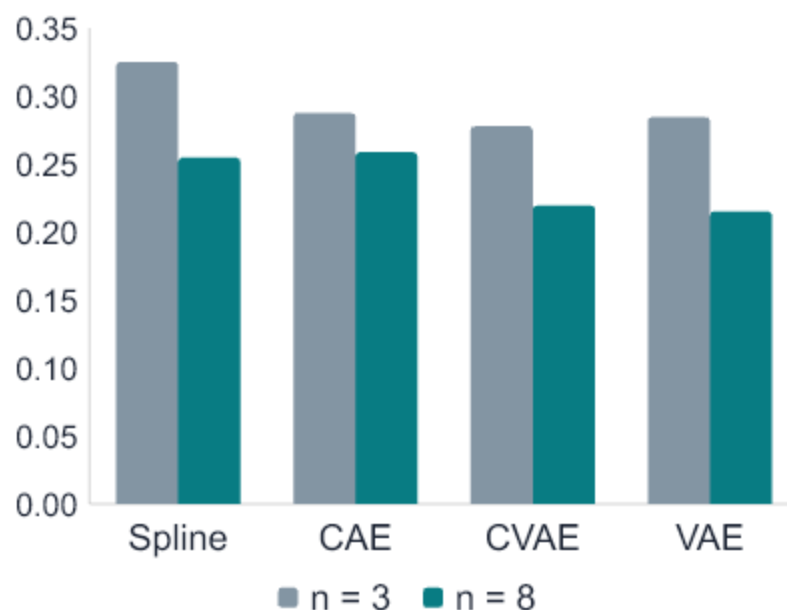
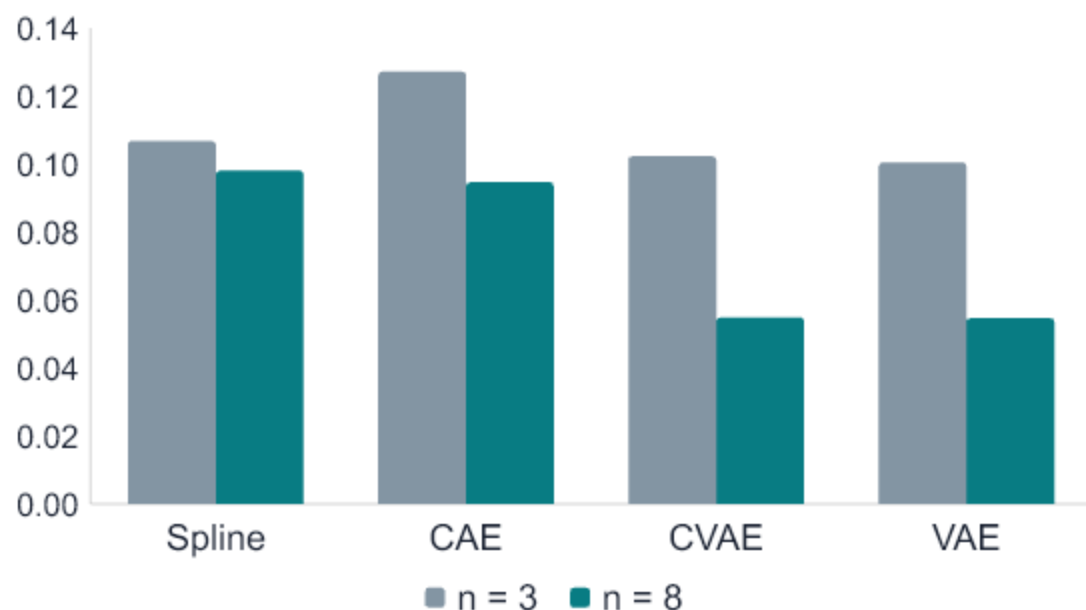


Illustration: subject01, n=8. VAE/CVAE beat spline on all three feature metrics at both capacities (BH and Holm-140). VAE and CVAE give comparable results.

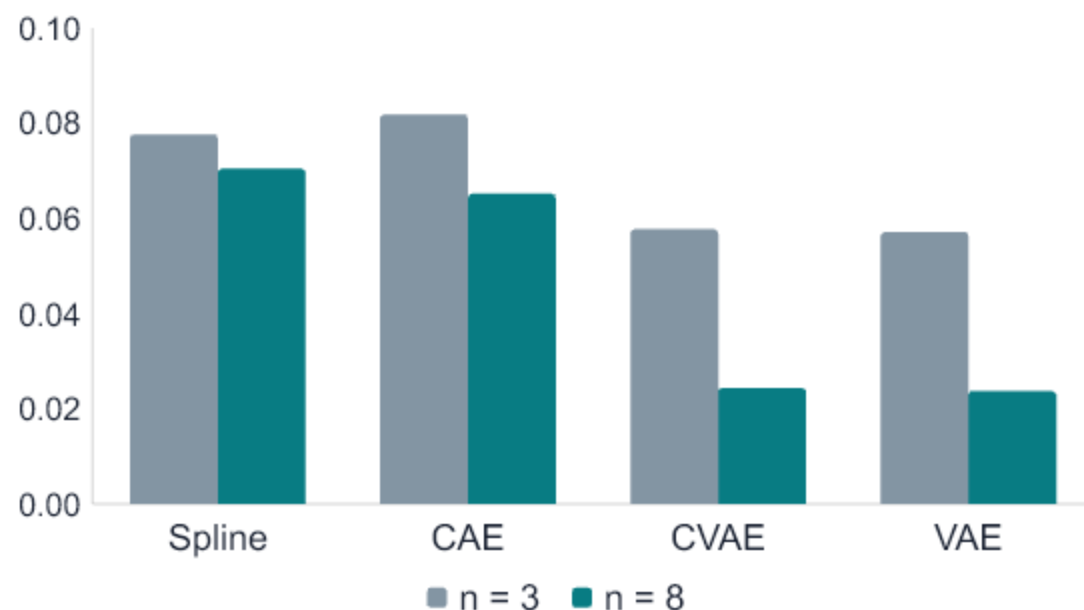
Full-trajectory generation supports VAE and CVAE

Energy and MMD compare complete sets of ordered paths, including their variability.

Full-path energy (cm)



Full-path MMD squared

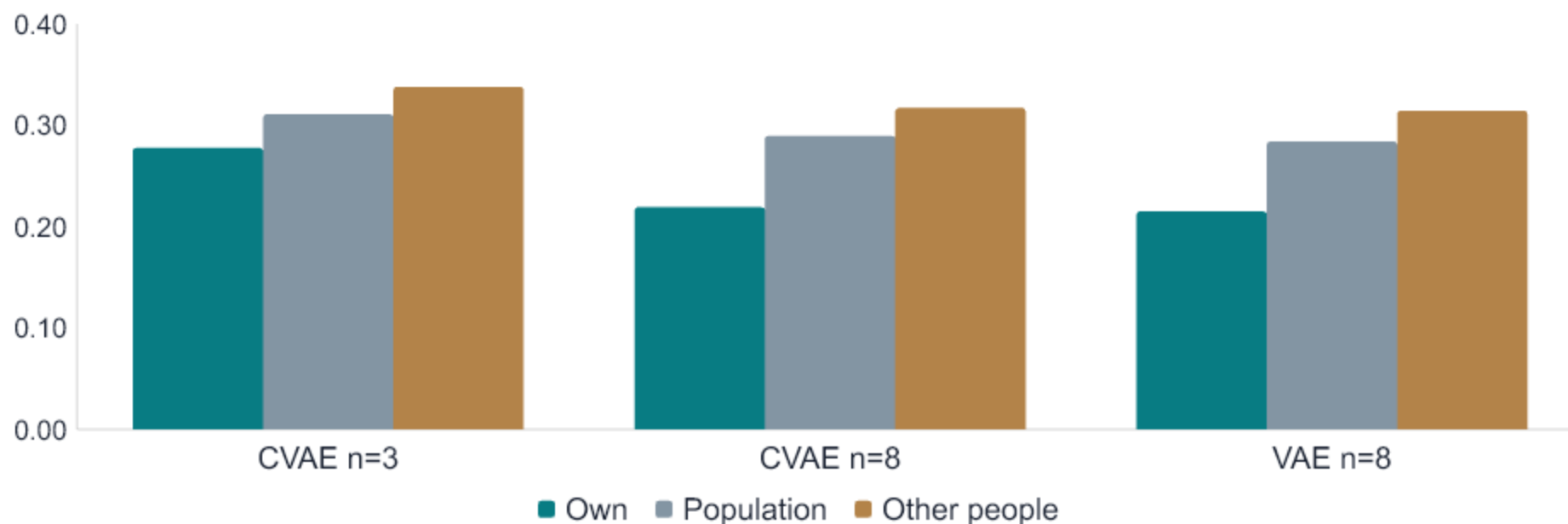


Lower is better. At n=8, both VAE/CVAE beat spline (BH and Holm-80), also with balanced axes. VAE vs CVAE: q=.672 and 1.000. At n=3, results depend on metric and scaling.

The participant's own center improves generation

We change the center while holding decoder, noise and task draws fixed.

Mean feature KS after replacing the center



Mean feature KS; lower is better. All VAE control comparisons: BH $q < 1.6 \times 10^{-6}$ (6 tests). All CVAE comparisons: $q < .000382$ (12 tests).

What the compact model achieves

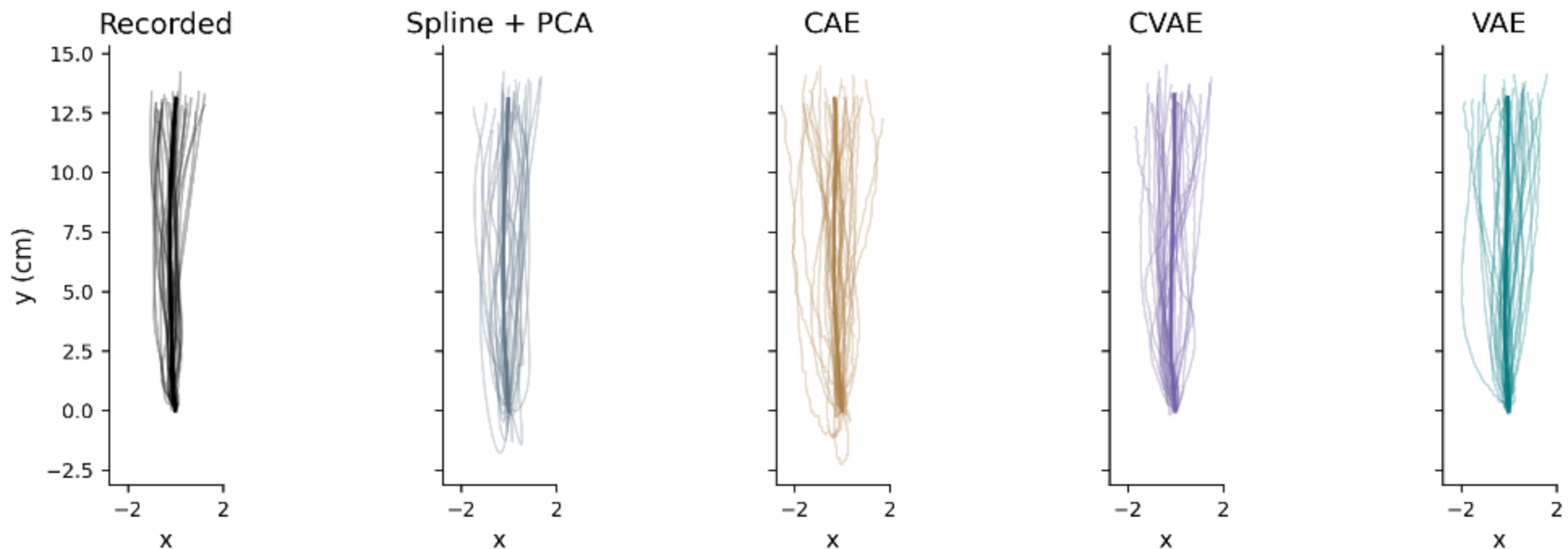
Stored context resampling has lower mean path discrepancy.
It retains complete paths, while the compact models retain a personal center.

Raw path reference	Energy (cm)	MMD ²
VAE n=8	0.0548	0.0238
Resample stored context	0.0328	0.0054
Repeat context mean	0.4621	0.2452

Descriptive reference means. Resampling retains all context paths; repeating their mean loses movement variability.

Dashboard demonstration

VAE $n=8$ is the default. Explore $n=3$, CVAE, CAE and spline + PCA alongside the cohort results.



Explore recorded and generated paths, feature distributions and model benchmarks for the same session.